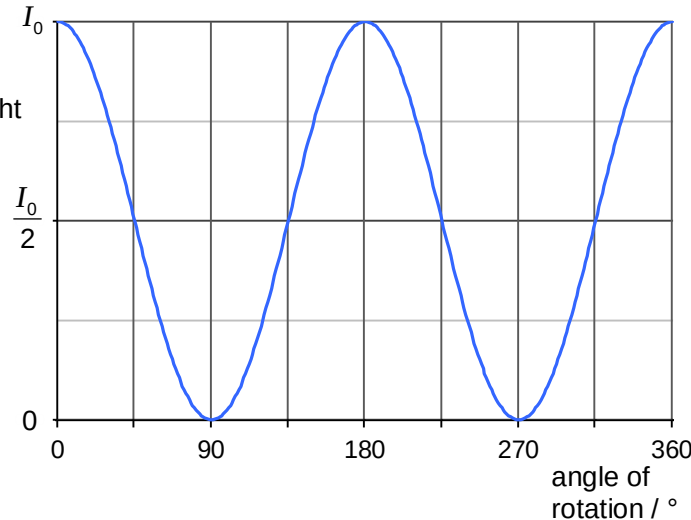


	$\approx 280 \text{ N}$
2(a)(i)	A transverse wave is a wave in which the oscillation of the particles is perpendicular to the propagation of the wave energy. An example is electromagnetic wave.
2(a)(ii)	Plane polarisation of a wave is the restriction of the oscillation of a transverse wave to one plane containing its direction of propagation and a single axis of oscillation perpendicular to it.

No.	Solution
2(b)(i)	Using Malus' law, Intensity of beam emerging from first filter, $I_1 = I_0 \cos^2 30^\circ = 0.75I_0$ The transmission axis of the second filter is $30^\circ (= 60^\circ - 30^\circ)$ with respect to the first filter. Intensity of beam emerging from second filter, $I_T = I_1 \cos^2 30^\circ$ $= (0.75I_0) \cos^2 30^\circ$ $= 0.5625I_0$ $\approx 0.56I_0$
2(b)(ii)	Using $I \propto A^2$, $\frac{I_T}{I_0} = \left(\frac{A_T}{A_0} \right)^2$ ratio $\frac{A_T}{A_0} = \sqrt{\frac{I_T}{I_0}}$ $= \sqrt{0.5625}$ $= 0.75$
2(b)(iii)	intensity of transmitted light  angle of rotation / °
2(c)(i)	Sound waves are longitudinal waves where the particles oscillate parallel to the direction of propagation of the wave energy and there is no oscillation that is perpendicular to its energy propagation. So it is not possible to restrict its oscillation to an axis perpendicular to its energy propagation. Thus sound waves cannot be polarised.
2(c)(ii)	Period $T = \frac{1}{f} = \frac{1}{740} = 1.351 \times 10^{-3} = 1.351 \text{ ms}$,

No.	Solution
	Time interval between two adjacent peaks = T Time-base setting on the oscilloscope $= \frac{T}{6.8} = \frac{1.351}{6.8}$ $= 0.1987$ $= 0.20 \text{ ms cm}^{-1}$
3(a)	The electric potential at a point in an electric field is defined as the work done per unit positive charge in bringing a small test charge from infinity to